

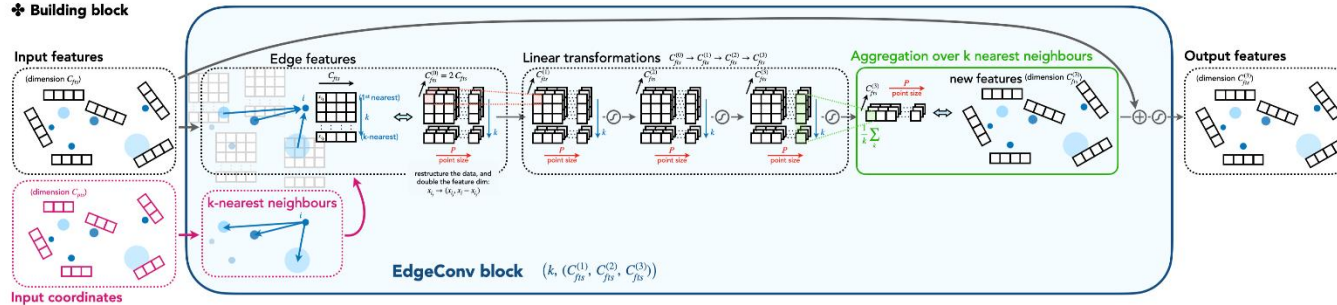
Mid-Project Report

Xinzhou Guo

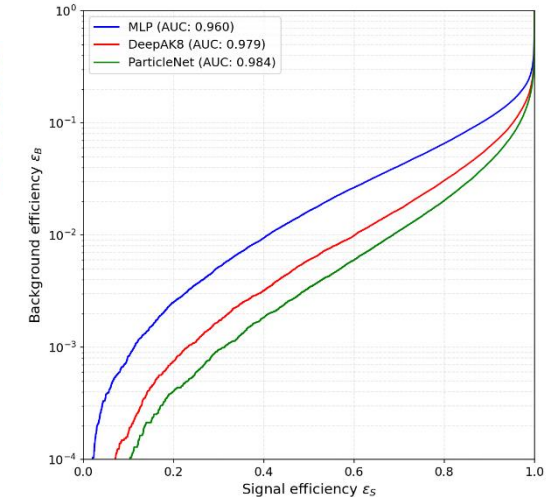
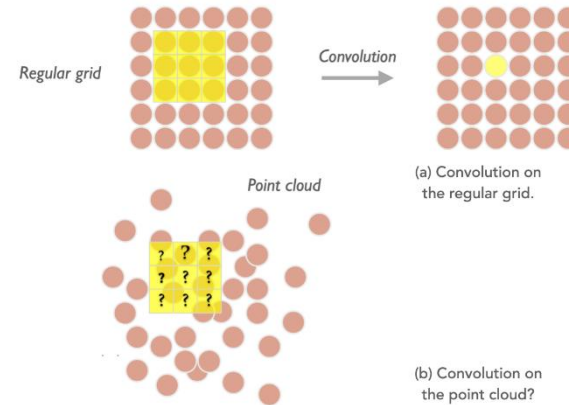
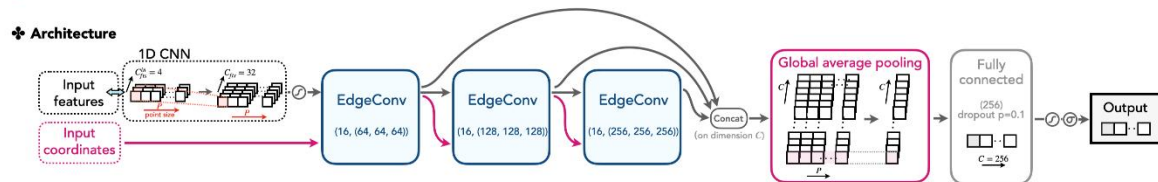
ParticleNet

- A graph neural net (GNN) model: treats candidates as point clouds
- The graph convolutional operation: EdgeConv & Dynamic Graph CNN
- Exploits correlations with k-nearest neighbors and generate latent graphs

Building block



Architecture



<https://cms-ml.github.io/documentation/inference/particlenet.html>

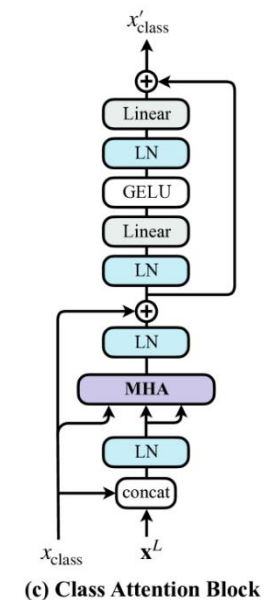
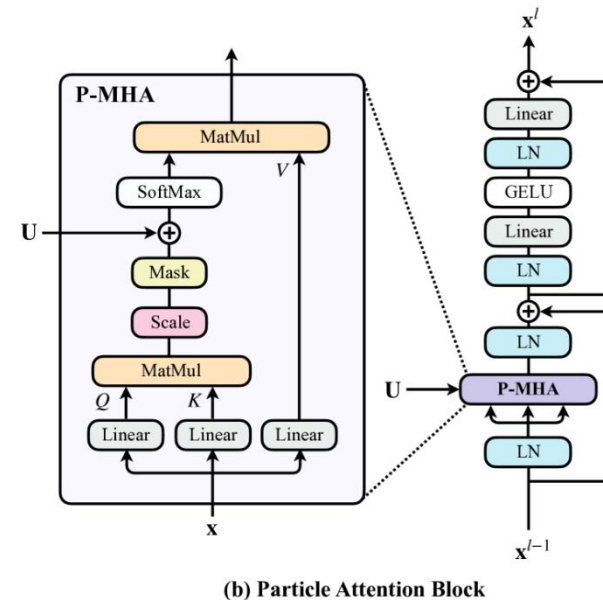
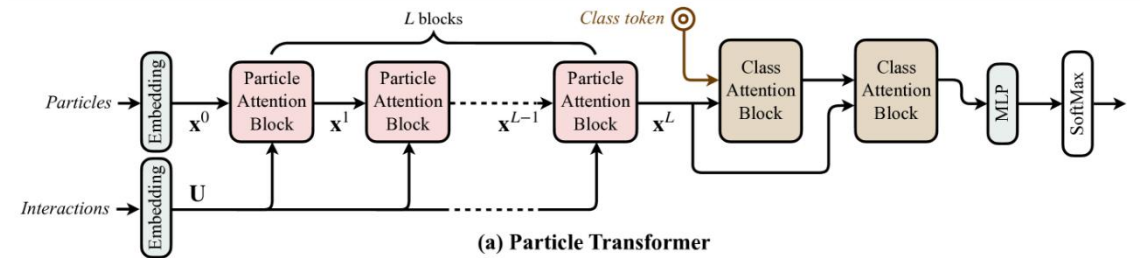
Particle Transformer (ParT)

- Pretraining

- Large dataset ($\sim 10\text{M}$ jets, many classes), used as a general feature extractor
- General-purpose jet representation, encodes rich jet features

- Fine-tuning

- Load pretrained GloParT
- Build analysis-specific dataset
- Train a lightweight classifier (MLP)
- Adapting to specific signal and/or background requirements for particular analyses



Current Progress

- Read the GloParT paper [Particle Transformer for Jet Tagging](#) and the documentation [Fine-tuning GloParT: Instructions](#)
- Feed the same dataset as ParticleNet or new datasets to train, compare results

MiniAOD to DNNTuples

To be continued

Fine-tune the full model with **weaver**

To be continued